AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 15 · WORKBOOK

# Source Quality

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

## One connected source-quality record.

Explain the policy,trace versions,audit missing precedence and plan a revealing check.

**Keep one habit:** Retain issuer,source origin,version links,query date and effective date together.

MAIN LESSON · SECTION 1

## Recall: reconnect earlier ideas

> Retrieve before opening notes.

### Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

**My remembered idea, example and correction**

**My remembered idea, example and correction**

▼ **Earlier wording, alternatives and picture transcripts**

**Hallucination Failure Types**

**Speech and Audio Systems**

**Prompt Evaluation**

## W1 · Explain source applicability

> Name query time,issuer authority and version relationship.

### Your evidence record

- Weekday review: N assigns room A from Monday;M is a signed amendment referencing N,room B effective Wednesday. Numbered-day case: A assigns Hall 1 fromday1;B changes toHall 3 fromday8;uploading old A on day 10 does not revoke B. Changedcase C explicitly revokesB fromday12,restoringHall 1. These are declared fictional policies;keep issuer,reference,effective date and query date separate.

- Reference for this exercise: open this week's student lesson, "Source Quality", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 13

## W2 · Trace the weekday amendment

> Do not substitute upload order for effective time.

### Your evidence record

- Weekday review: N assigns room A from Monday;M is a signed amendment referencing N,room B effective Wednesday. Numbered-day case: A assigns Hall 1 fromday1;B changes toHall 3 fromday8;uploading old A on day 10 does not revoke B. Changedcase C explicitly revokesB fromday12,restoringHall 1. These are declared fictional policies;keep issuer,reference,effective date and query date separate.

- Notice N says room A from Monday. Signed amendment M references N and changes to B effective Wednesday. Query asks Tuesday, then Thursday. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

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MAIN LESSON · SECTION 14

## W3 · Audit an absent precedence rule

> An unspecified authority policy cannot be invented.

### Your evidence record

- Weekday review: N assigns room A from Monday;M is a signed amendment referencing N,room B effective Wednesday. Numbered-day case: A assigns Hall 1 fromday1;B changes toHall 3 fromday8;uploading old A on day 10 does not revoke B. Changedcase C explicitly revokesB fromday12,restoringHall 1. These are declared fictional policies;keep issuer,reference,effective date and query date separate.

- A report omits this qualification: An unspecified **authority** or effective-time rule leaves a conflict unresolved; do not invent precedence.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 16

# W5 · Design a revealing source check

> Preserve source origins and the expected time-scoped answer.

## Your evidence record

- Weekday review: N assigns room A from Monday;M is a signed amendment referencing N,room B effective Wednesday. Numbered-day case: A assigns Hall 1 fromday1;B changes toHall 3 fromday8;uploading old A on day 10 does not revoke B. Changedcase C explicitly revokesB fromday12,restoringHall 1. These are declared fictional policies;keep issuer,reference,effective date and query date separate.

- Reference for this exercise: open this week’s student lesson, “Source Quality”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

MAIN LESSON · SECTION 4

## W4 · Trace a new capacity version independently

> Keep one first attempt before hints or feedback.

### Read and explain independently

Apply the supplied classroom amendment policy,including its effective-day boundary. Use only the original and signed amendment described in this task,and retain their relationship. No real room inspection,extra version or new authority rule is supplied.

Original allows capacity 14. Signed amendment lowers it to 9 effective day 5. Query concerns days 4 and 6. Trace both and preserve versions.

**My independent explanation, trace and limits**

## Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

**Independent case: Apply a new case independently**

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MAIN LESSON · SECTION 9

# Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

## Your evidence record

Use an approved source list and record why each source fits the learner's question.

The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

## Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

### Your evidence record

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

# Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

## Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

### Use this when helpful

Use W2: Trace the worked case for Conflicting and time-sensitive sources. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

## Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

### Use this when helpful

Use the worked case for Conflicting and time-sensitive sources. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

## Reference · Original learning target

> Use this target to inspect an explanation.

### Use this when helpful

A school notice updated Friday conflicts with a blog quoting Monday's notice about a room change. Which source should you inspect first, and why is newest alone not enough?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

## Extra practice · A complete case

> Try the question before checking the worked explanation.

### Use this when helpful

An original experiment report and an anonymous unlinked summary describe a method.

Where should verification begin?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

## Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

### Use this when helpful

For a school's current opening time, compare an undated repost with a dated notice issued by the school. Which is the stronger starting source, and what must still be checked?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

**Week 15 Test and reflect**

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

## Extra practice · Design a revealing test

> Predict before tracing or testing.

### Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

▼ Reference · Original target response space

REFERENCE · SECTION 11

# Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

## Use this when helpful

A school notice updated Friday conflicts with a blog quoting Monday's notice about a room change. Which source should you inspect first, and why is newest alone not enough?

### My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

# Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

## Use this when helpful

Original allows capacity 14. Signed amendment lowers it to 9 effective day 5. Query concerns days 4 and 6. Trace both and preserve versions.

**My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

# Extra practice · Use the worked picture

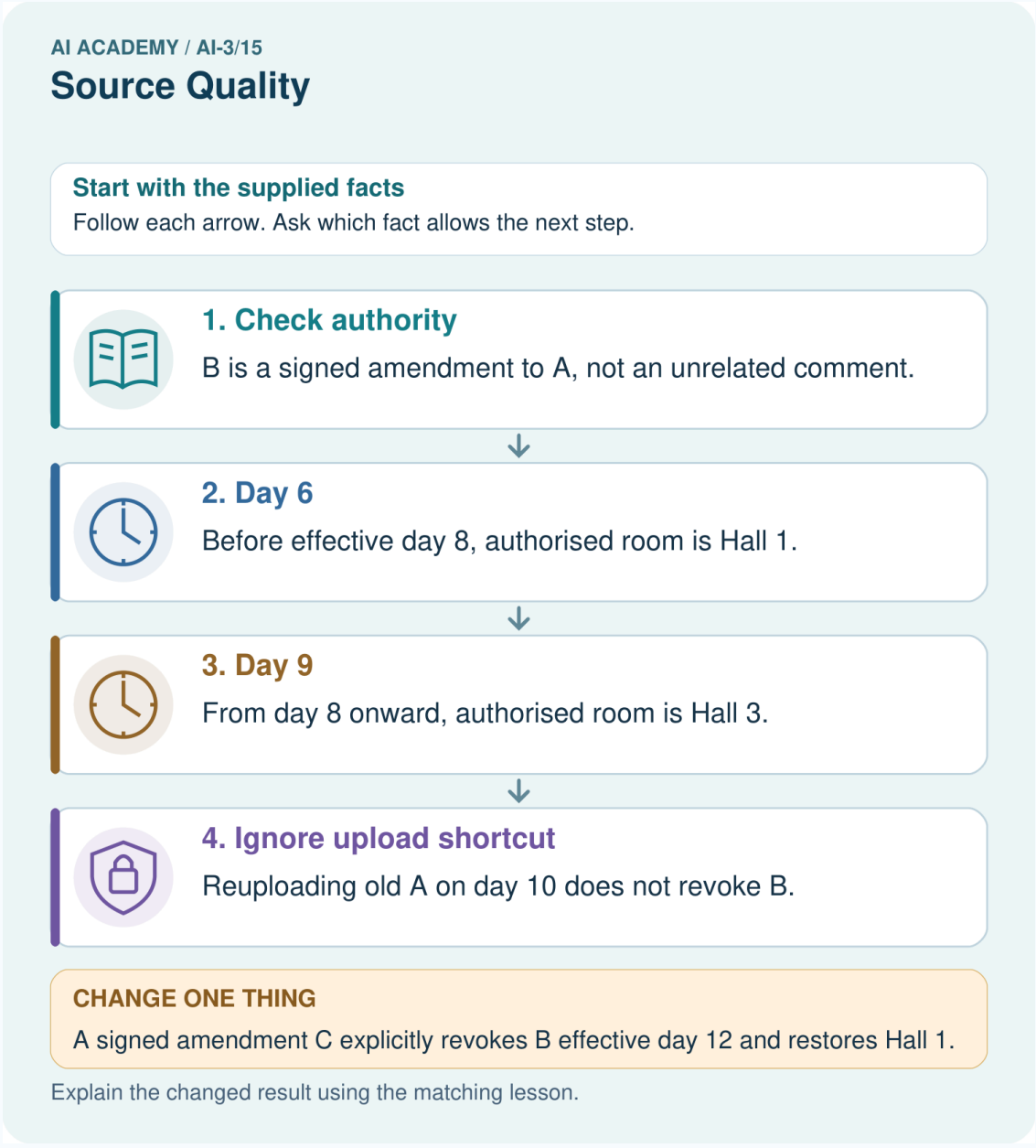
> Keep the explanation and the support you used.

## Use this when helpful

A fictional Study Buddy answers which room a club is authorised to use.

Signed notice A: Hall 1 from day 1. Signed amendment B references A and changes to Hall 3 effective day 8.

An old copy of A is reuploaded on day 10. B remains the authoritative amendment. Queries concern day 6 and day 9.




Follow the labelled picture with the matching explanation. Open the vector diagram

Which source fields must be checked before answering day 9?

**My picture-based explanation and changed-case prediction**

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Store Study Buddy source identity, version, **authority** and effective date, especially for schedules or changing classroom rules.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

# Reference · Project portfolio record

> Extend your main project response when helpful.

## Use this when helpful

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Store Study Buddy source identity, version, **authority** and effective date, especially for schedules or changing classroom rules.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |

| Evidence to keep | My record |
| --- | --- |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

# Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

## First supported question

- A fictional Study Buddy answers which room a club is authorised to use.
- Which labelled step matches this detail? Reuploading old A on day 10 does not revoke B.
- 1. Day 9
- 2. Ignore upload shortcut
- 3. Day 6

- 4. Check **authority**

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Ignore upload shortcut Reuploading old A on day 10 does not revoke B. A source's upload time and the time its rule takes effect answer different questions.

Preserve version relationships and the query's intended date. The current authoritative chain gives a consistent answer to both queries.

A later filename or timestamp alone cannot establish that an old record supersedes a signed amendment.

## Changed-case question

- Changed-case practice: A signed amendment C explicitly revokes B effective day 12 and restores Hall 1. Which conclusion is supported by the supplied facts?

- 1. Day 9 stays Hall 3; day 13 becomes Hall 1.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Day 9 stays Hall 3; day 13 becomes Hall 1. The new authoritative change applies from its stated effective date, not retroactively.

## Repair and save your work

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

Review Speech and Audio Systems (ai-3/12). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?. Return to this week and explain what you changed.

Review Prompt Evaluation (ai-3/9). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Why is B′s selection defensible here without calling it universally superior?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

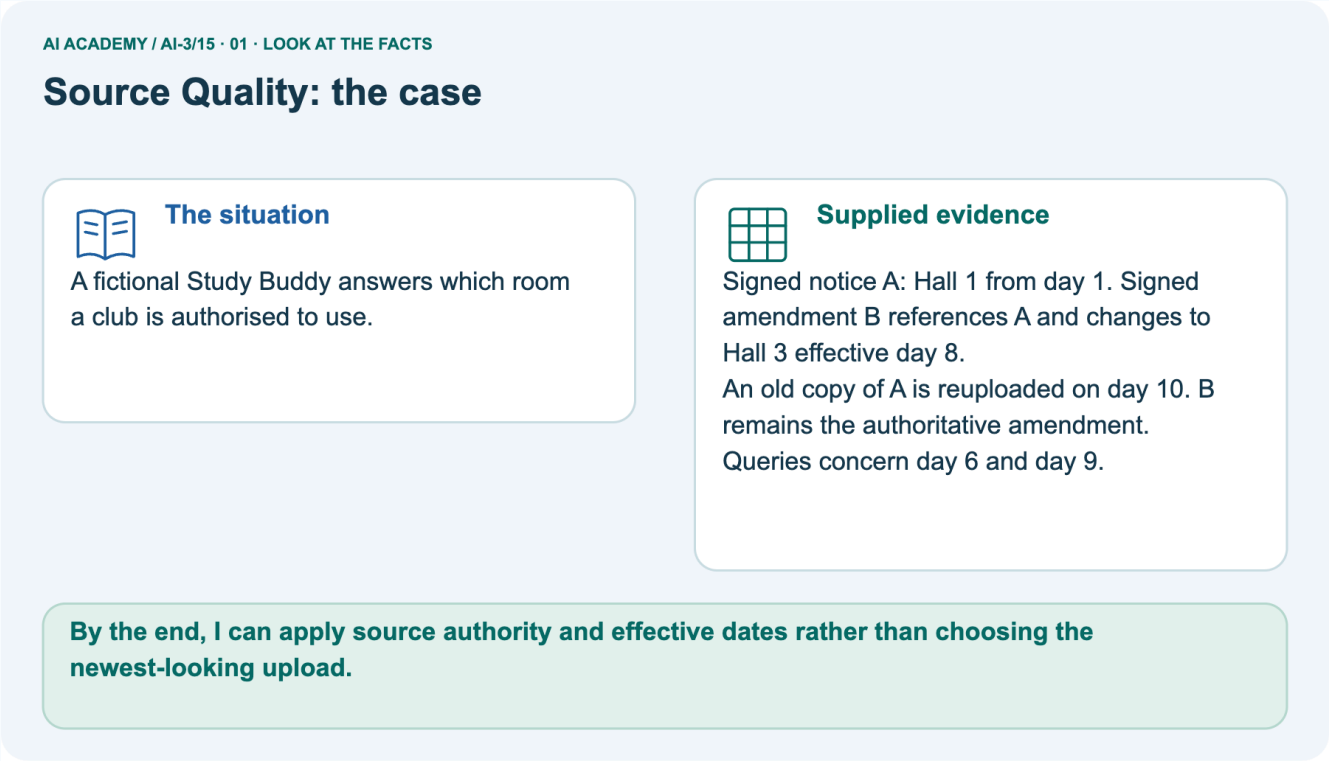
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

## Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

## 01. Source Quality: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

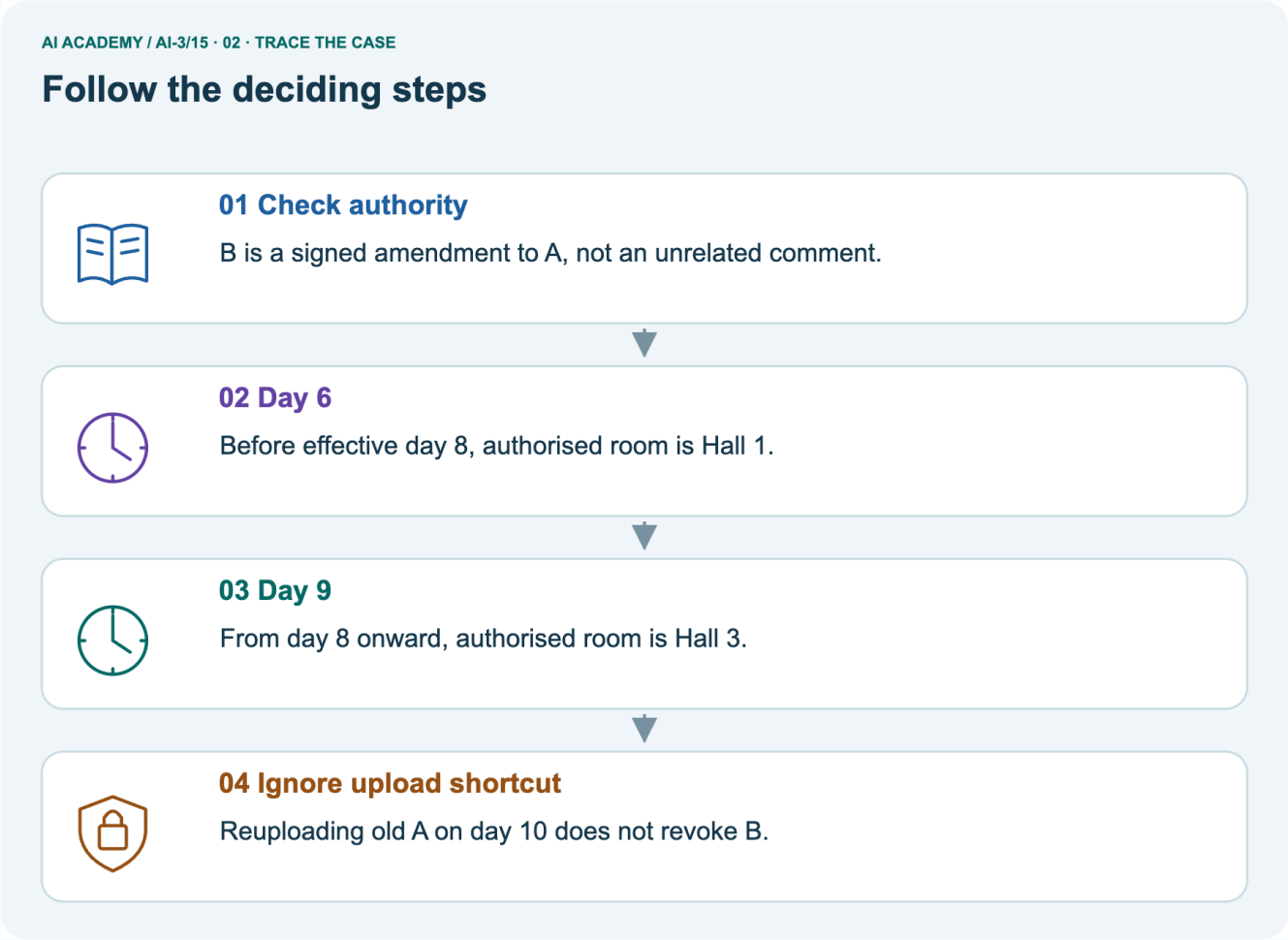
A fictional Study Buddy answers which room a club is authorised to use. Supplied facts: Signed notice A: Hall 1 from day 1. Signed amendment B references A and changes to Hall 3 effective day 8.

An old copy of A is reuploaded on day 10. B remains the authoritative amendment. Queries concern day 6 and day 9.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



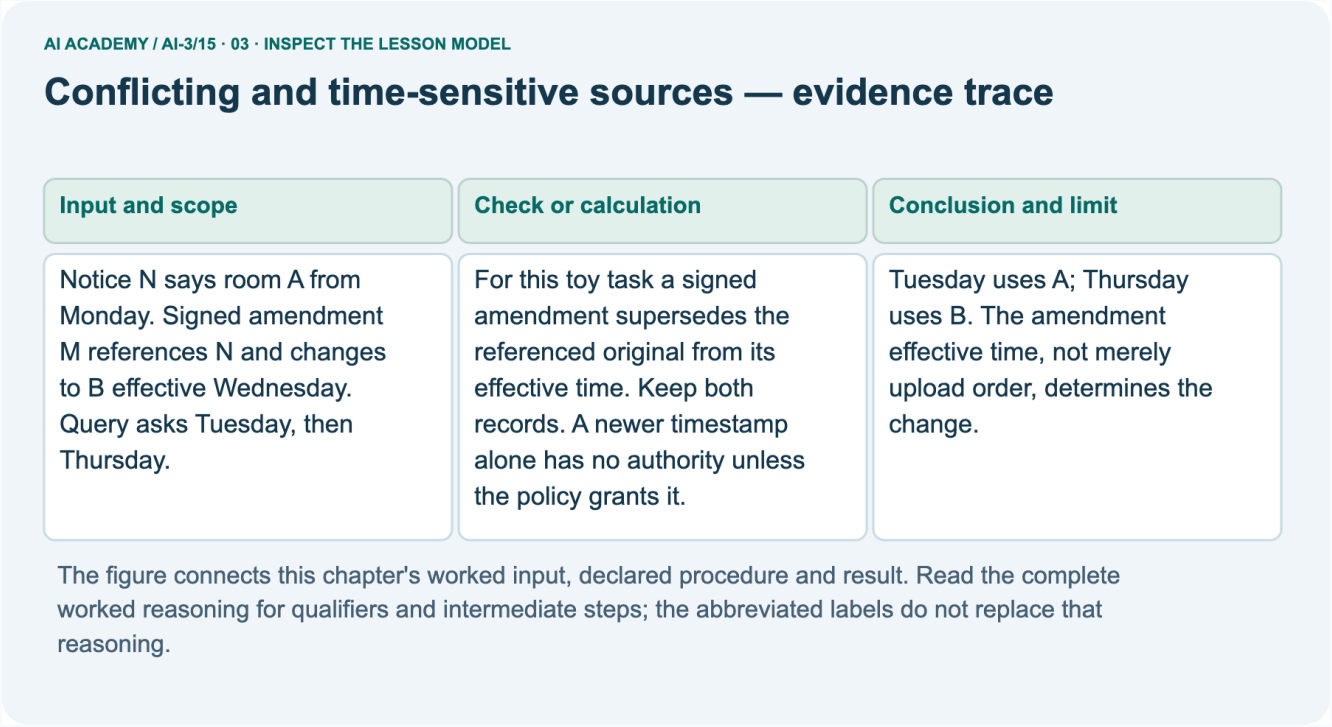

Follow the labelled picture with the matching explanation. Open the vector diagram

Check **authority**: B is a signed amendment to A, not an unrelated comment. → Day 6: Before effective day 8, authorised room is Hall 1. → Day 9: From day 8 onward, authorised room is Hall 3. → Ignore upload shortcut: Reuploading old A on day 10 does not revoke B.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Conflicting and time-sensitive sources — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

Conflicting and time-sensitive sources — **evidence** trace. Notice N says room A from Monday. Signed amendment M references N and changes to B effective Wednesday.

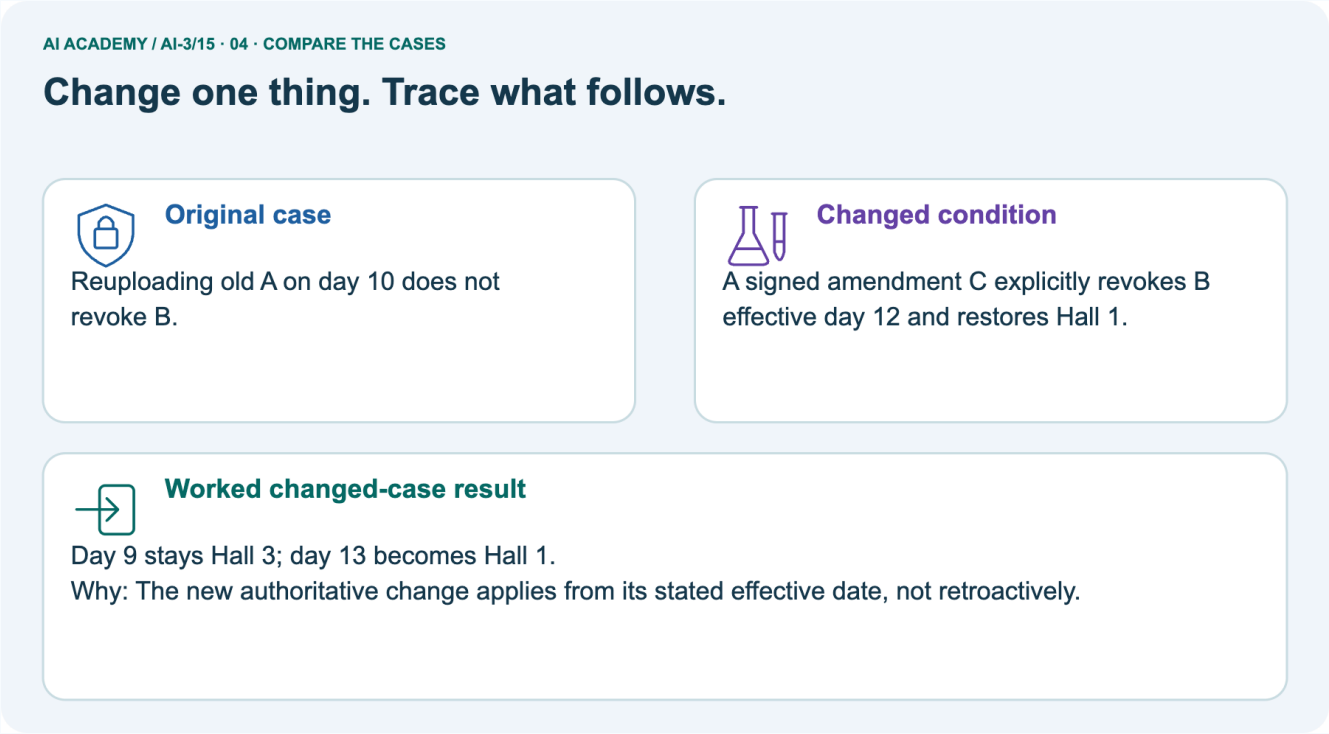
Query asks Tuesday, then Thursday. | For this toy task a signed amendment supersedes the referenced original from its **effective time**. Keep both records.

A newer timestamp alone has no **authority** unless the policy grants it. | Tuesday uses A; Thursday uses B. The amendment **effective time**, not merely upload order, determines the change.

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

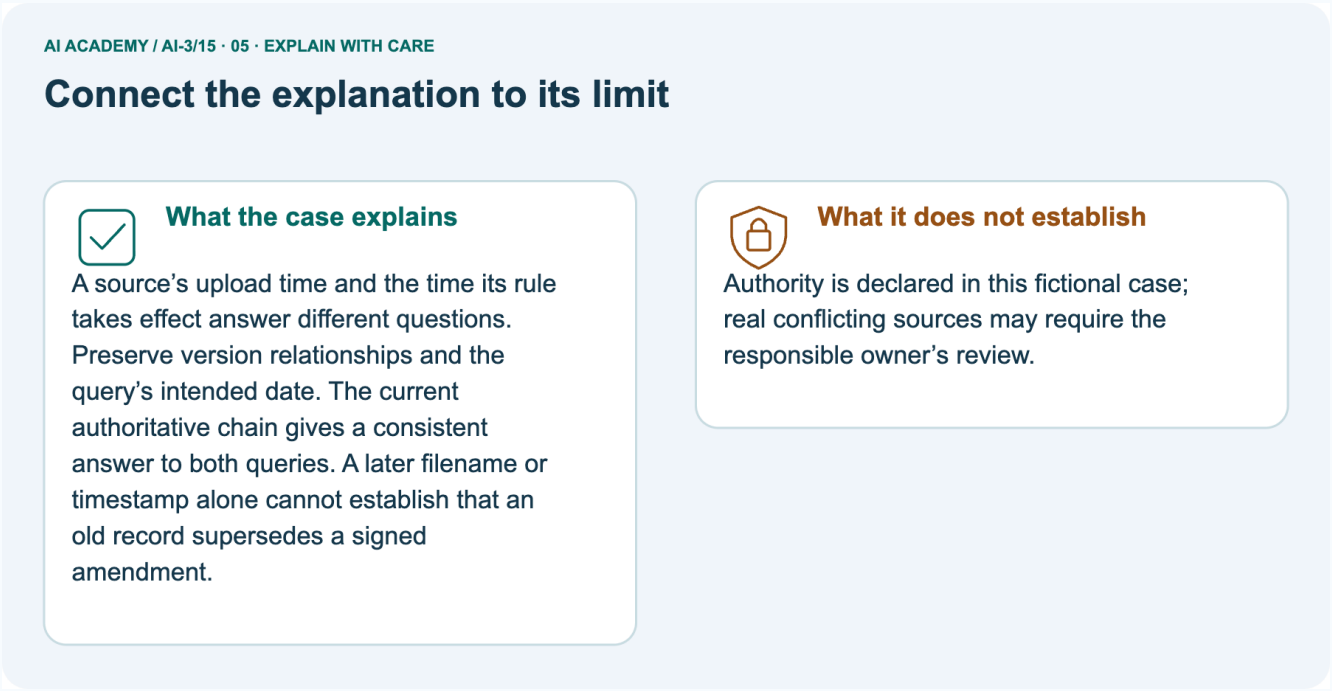
Original case: Reuploading old A on day 10 does not revoke B. Changed condition: A signed amendment C explicitly revokes B effective day 12 and restores Hall 1. Expected result: Day 9 stays Hall 3; day 13 becomes Hall 1.

Reason: The new authoritative change applies from its stated effective date, not retroactively.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

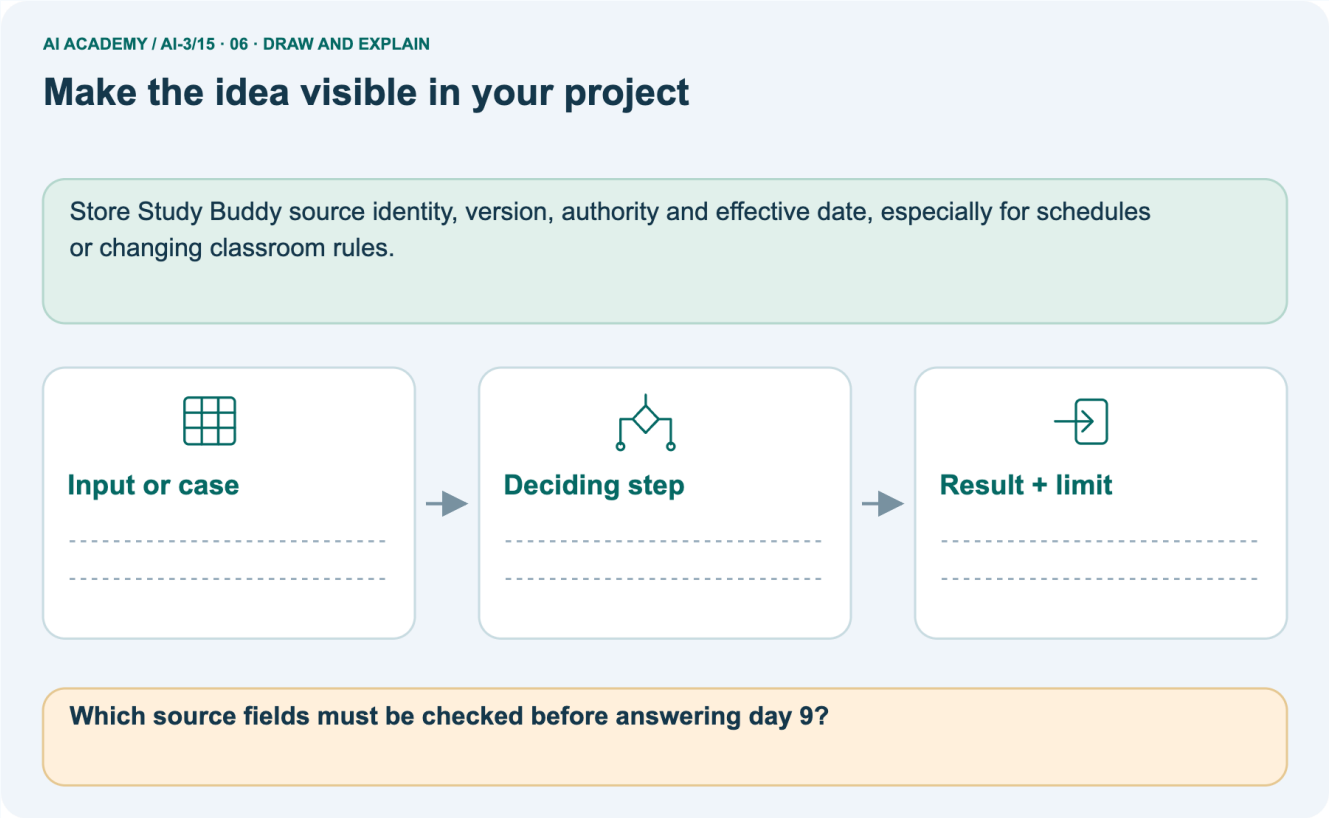
A source's upload time and the time its rule takes effect answer different questions. Preserve version relationships and the query's intended date. The current authoritative chain gives a consistent answer to both queries.

A later filename or timestamp alone cannot establish that an old record supersedes a signed amendment. Limit: **Authority** is declared in this fictional case; real conflicting sources may require the responsible owner's review.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Store Study Buddy source identity, version, **authority** and effective date, especially for schedules or changing classroom rules. Draw your **input**, deciding step, result and limit.

Supported question: Which source fields must be checked before answering day 9?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

**01. Source Quality: the case**

**02. Follow the deciding steps**

**03. Conflicting and time-sensitive sources — evidence trace**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.